

A Practical Guide to Support vector machine

TOPIC -- SUPPORT VECTOR MACHINE

-- 01 --

$$\left[\frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i (\mathbf{w}^T \mathbf{x} - b)) \right] + \lambda \|\mathbf{w}\|^2. \quad \{\displaystyle \left[\frac{1}{n}\sum_{i=1}^n \max \left(0, 1 - y_i (\mathbf{w}^T \mathbf{x} - b)\right)\right] + \lambda \|\mathbf{w}\|^2. \}$$

-- 02 --

SVMs are helpful in text and hypertext categorization, as their application can significantly reduce the need for labeled training instances in both the standard inductive and transductive settings. Some methods for shallow semantic parsing are based on support vector machines. Classification of images can also be performed using SVMs. Experimental results show that SVMs achieve significantly higher search accuracy than traditional query refinement schemes after just three to four rounds of relevance feedback. This is also true for image segmentation systems, including those using a modified version SVM that uses the privileged approach as suggested by Vapnik. Classification of satellite data like SAR data using supervised SVM. Hand-written characters can be recognized using SVM. The SVM algorithm has been widely applied in the biological and other sciences. They have been used to classify proteins with up to 90% of the compounds classified correctly. Permutation tests based on SVM weights have been suggested as a mechanism for interpretation of SVM models. Support vector machine weights have also been used to interpret SVM models in the past. Posthoc interpretation of support vector machine models in order to identify features used by the model to make predictions is a relatively new area of research with special significance in the biological sciences.

-- 03 --

$$\begin{aligned} \text{minimize } & \|\mathbf{w}\|^2 + C \sum_{i=1}^n \zeta_i \text{ subject to } y_i (\mathbf{w}^T \mathbf{x}_i - b) \geq 1 - \zeta_i, \\ & \zeta_i \geq 0 \quad \forall i \in \{1, \dots, n\} \end{aligned} \quad \{\displaystyle \begin{aligned} \text{minimize } & \|\mathbf{w}\|^2 + C \sum_{i=1}^n \zeta_i \text{ subject to } y_i (\mathbf{w}^T \mathbf{x}_i - b) \geq 1 - \zeta_i, \\ & \zeta_i \geq 0 \quad \forall i \in \{1, \dots, n\} \end{aligned} \}$$

-- 04 --

Properties SVMs belong to a family of generalized linear classifiers and can be interpreted as an extension of the perceptron. They can also be considered a special case of Tikhonov regularization. A special property is that they simultaneously minimize the empirical classification error and maximize the geometric margin; hence they are also known as maximum margin classifiers. A comparison of the SVM to other classifiers has been made by Meyer, Leisch and Hornik.

-- 05 --

$$\min_{\mathbf{w}} \left\| \mathbf{w} \right\|^2 + C \left[\frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i (\mathbf{w}^T \mathbf{x}_i - b)) \right],$$

-- 06 --

$\mathbf{w}^T \mathbf{x} - b = -1$ (anything on or below this boundary is of the other class, with label -1). Geometrically, the distance between these two hyperplanes is $\frac{2}{\|\mathbf{w}\|}$, so to maximize the distance between the planes we want to minimize $\|\mathbf{w}\|$. The distance is computed using the distance from a point to a plane equation. We also have to prevent data points from falling into the margin, we add the following constraint: for each i either

-- 07 --

This is called the dual problem. Since the dual maximization problem is a quadratic function of the c_i subject to linear constraints, it is efficiently solvable by quadratic programming algorithms. Here, the variables c_i are defined such that

-- 08 --

Note that y_i is the i -th target (i.e., in this case, 1 or -1), and $\mathbf{w}^T \mathbf{x}_i - b$ is the i -th output. This function is zero if the constraint in (1) is satisfied, in other words, if $\mathbf{x}_i$ lies on the correct side of the margin. For data on the wrong side of the margin, the function's value is proportional to the distance from the margin. The goal of the optimization then is to minimize:

-- 09 --

Risk minimization In supervised learning, one is given a set of training examples $X_1 \dots X_n$ with labels $y_1 \dots y_n$

$y_1 \dots y_n$, and wishes to predict y_{n+1} given X_{n+1} . To do so one forms a hypothesis, f , such that $f(X_{n+1})$ is a "good" approximation of y_{n+1} . A "good" approximation is usually defined with the help of a loss function, $\ell(y, z)$, which characterizes how bad z is as a prediction of y . We would then like to choose a hypothesis that minimizes the expected risk:

-- 10 --

For the square-loss, the target function is the conditional expectation function, $f_{sq}(x) = \mathbb{E}[y|x]$; For the logistic loss, it's the logit function, $f_{log}(x) = \ln \frac{p_x}{1-p_x}$. While both of these target functions yield the correct classifier, as $\text{sgn}(f_{sq}) = \text{sgn}(f_{log}) = f^*$, they give us more information than we need. In fact, they give us enough information to completely describe the distribution of y_x . On the other hand, one can check that the target function for the hinge loss is exactly f^* . Thus, in a sufficiently rich hypothesis space—or equivalently, for an appropriately chosen kernel—the SVM classifier will converge to the simplest function (in terms of R) that correctly classifies the data. This extends the geometric interpretation of SVM—for linear classification, the empirical risk is minimized by any function whose margins lie between the support vectors, and the simplest of these is the max-margin classifier.